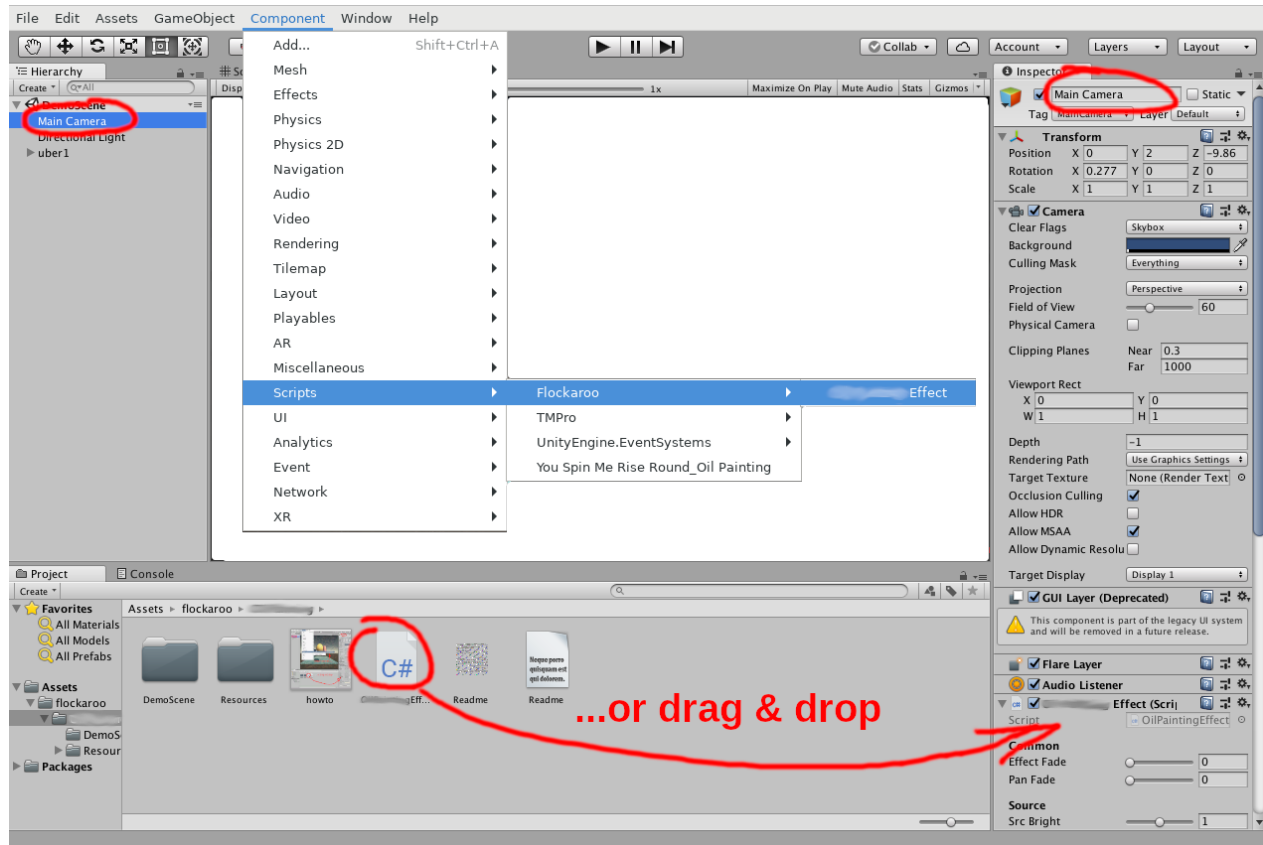


Aquarelle - Unity3D Image Effect

(c) 2020 by [flockaroo](#) (Florian Berger) - email: flockaroo@gmail.com

How to use

Select your camera node and then simply add "Aquarelle" script to camera components (can be found in Assets/flockaroo/Aquarelle/). You can drag/drop it to there or choose it from the menu (Component/Scripts/Flockaroo/Aquarelle).



Warning!! The subfolder "flockaroo_Aquarelle" in "Resources" is needed by the effect script for unique identification of files and should not be removed or renamed.

Parameters

The shader provides the following parameters:

Input/Output

Parameter	function
Input Texture	take this texture as input instead of the camera
Render To Texture	render to texture instead of screen
Output Texture	texture being rendered to if above is checked
Output Mipmap	generate mipmap for output texture

Source Adjustments

Parameter	function
Src Brightness	adjust source brightness
Src Contrast	adjust source contrast
Src Color	adjust source color

Effect

Parameter	function
Linear2Gamma	the effect was originally made for gamma colorspace. if you use linear space (default in recent unity) check this box to get better results
Master Fade	1 = effect image ... 0 = original content
Content Vign Aspect	vignette aspect setting
Content Vign Sharp	content vignette sharpness
Content Vign Size	content vignette size
Paper Rough	paper roughness
Predraw Strength	predraw strength
Predraw Amount	predraw number of lines (0=none 1=many)
FastGrad	optimized for faster render (but looks slightly different)
Num Samples	number of samples 2..64 (less samples -> faster but less quality)
Spread	adjust how much aquarelle color spreads
Color Spread	add variation to color spots
StrokeScale	size of color spots
Rim Dry	color spots darker on drying edge
Vignette	strength of vignette

Parameter	function
Mask Texture	texture to mask the effect (0 no effect 1 full effect)
UseMask	use Mask Texture [0..1]

Other

Parameter	function
Flip Y	image Y flip
Geom Flip Y	Y-flip of effect-internal geometry (use this if "Effect Fade" and "Pan Fade" wont work properly)
HDRP Gamma	check this if you are using linear color space (only active in hdrp mode)

concerning "Flip Y" and "Geom Flip Y":

The screen coordinates of unity are a bit mysterious. even more when working on different platforms. The Y-coordinate seems to be flipped between versions even on the same system, and also flipped depending on the system.

So for "Flip Y" and "Geom Flip Y" follow these rules:

If you have the source ("Effect Fade" to 1) flipped and the effect correct, just check "Geom Flip Y".

If you have the source correct and the effect flipped, check both "Geom Flip Y" and "Flip Y".

If both are equally flipped just check "Flip Y".

Color Spaces

All flockaroo-effects are initially designed for Gamma-Space. Gamma-space was the default in earlier versions. At first HDRP came up with using HDR-Color space, which is a linear colorspace. So in HDRP and URP versions there's an additional parameter named "HDRP Gamma", which should compensate for washing out contrasts in linear colorspace. However in later versions linear color space is also being used in the Standard render-pipeline. So all versions (std/URP/HDRP) have now an additional parameter "linear2Gamma", which should be enabled as soon as you are using linear color space. As for today the "HDRP Gamma" parameter should be disabled in most cases. It can sometimes give better results e.g. in HDRP, but linear2Gamma should be disabled then. both being enabled won't make much sense in general.

HDRP (disabled by default)

The hdrp file is disabled by default !!! here's how to use it:

Unity wont compile this effect properly if no hdrp support is present on your version, so in the hdrp ".cs" file in the very first line the `"//USE_HDRP"` must be uncommented to make use the hdrp effect.

You also have to add it to the list of effects known to your project:

"Edit/Project Settings... -> HDRP Default Settings -> After Post Process"

..and then add it as an effect volume by clicking "Add Override" and the selecting

"Post-processing/Custom/Flockaroo/..."

from the menu.

URP (disabled by default)

The URP file is disabled by default !!! here's how to use it:

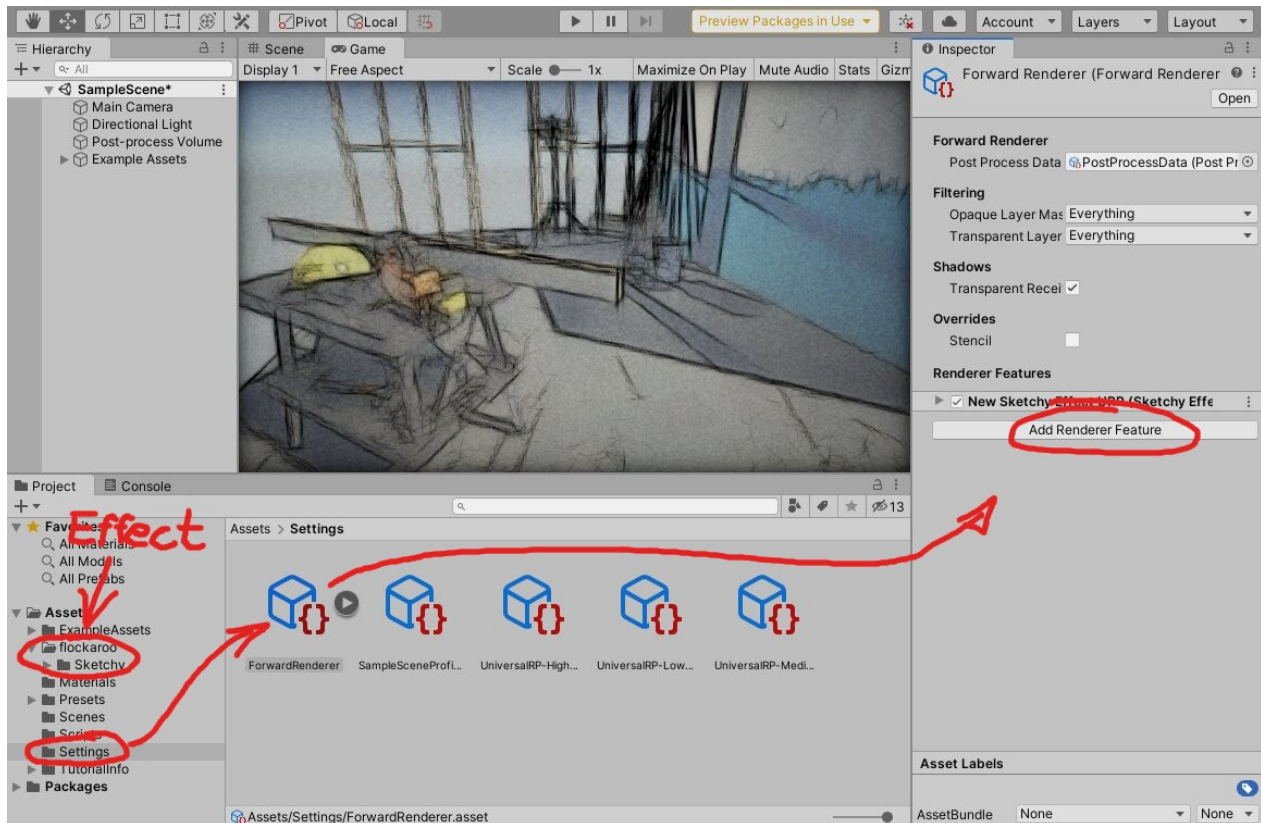
Unity wont compile this effect properly if no URP-support is present on your version, so in the urp "...URP.cs" file in the very first line the `"//USE_URP"` must be uncommented to make use the urp effect.

Then under "Assets/Settings/ForwardRenderer" press "Add Renderer Feature" in the Inspector Tab. In newer versions there are several URP Renderers URP Balanced, URP HighDynamic, URP Performant instead of Forward. Beware: This must be the same Renderer as is configured under "ProjectSettings/Quality/Rendering".

For older versions you might also want to comment the line `"#define URP_VERSION_GE_13"` for the effect to work.

It should also run in Unity 6, but rendergraph needs to be disabled by switching to compatibility mode

"Edit/ProjectSettings/Graphics/URP/RenderGraph"



BEWARE!! For now the effect can not be used after Post Processing.
Furthermore some Post-Processing-Effects like "Bloom" dont work properly. Disable those effects for proper functionality.